Linear Algebra and Its Applications

Section 6.1: Inner Product, Length, and Orthogonality

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

6.1 Inner product, length, and orthogonality 3

Practice problem 6.1.P1: A projection coefficient and its scaled vector 3

Practice problem 6.1.P2: Normalize the three-coordinate vector 3

Practice problem 6.1.P3: Check orthogonality directly 3

Practice problem 6.1.P4: Orthogonality survives normalization 3

Exercise 6.1.1: Inner-product and norm calculations 4

Exercise 6.1.2: Inner-product and norm calculations 4

Exercise 6.1.3: Inner-product and norm calculations 4

Exercise 6.1.4: Inner-product and norm calculations 4

Exercise 6.1.5: Inner-product and norm calculations 4

Exercise 6.1.6: Inner-product and norm calculations 5

Exercise 6.1.7: Inner-product and norm calculations 5

Exercise 6.1.8: Inner-product and norm calculations 5
Exercise 6.1.9: Normalize without changing the direction 5
Exercise 6.1.10: Normalize without changing the direction 5
Exercise 6.1.11: Normalize without changing the direction 6
Exercise 6.1.12: Normalize without changing the direction 6
Exercise 6.1.13: Inner-product and norm calculations 6
Exercise 6.1.14: Inner-product and norm calculations 7
Exercise 6.1.15: Inner-product and norm calculations 7
Exercise 6.1.16: Inner-product and norm calculations 7
Exercise 6.1.17: Inner-product and norm calculations 7
Exercise 6.1.18: Inner-product and norm calculations 7
Exercise 6.1.19: Inner-product and norm calculations 8
Exercise 6.1.20: Inner-product and norm calculations 8
Exercise 6.1.21: Inner-product and norm calculations 8
Exercise 6.1.22: Inner-product and norm calculations 8
Exercise 6.1.23: Inner-product and norm calculations 9
Exercise 6.1.24: Inner-product and norm calculations 9
Exercise 6.1.25: Inner-product and norm calculations 9
Exercise 6.1.26: Inner-product and norm calculations 9
Exercise 6.1.27: Inner-product and norm calculations 9
Exercise 6.1.28: Inner-product and norm calculations 10
Exercise 6.1.29: Inner-product and norm calculations 10
Exercise 6.1.30: Inner-product and norm calculations 10
Exercise 6.1.31: Inner-product and norm calculations 10
Exercise 6.1.32: Inner-product and norm calculations 10
Exercise 6.1.33: Inner-product and norm calculations 11
Exercise 6.1.34: Inner-product and norm calculations 12

6.1 Inner product, length, and orthogonality

For real columns u, v of equal length, $u \cdot v = u^T v$ is the sum of corresponding coordinate products. The norm is $\|v\| = \sqrt{v \cdot v}$, and distance is $\|u - v\|$. Two vectors are orthogonal precisely when their inner product is zero. Normalize a nonzero vector by dividing by its length, not by its squared length. The zero vector has no unit vector in its direction.

For a subspace W, its orthogonal complement $W^\perp$ contains the vectors orthogonal to *every* vector of W. Testing a spanning family suffices because the inner product distributes over linear combinations.

Practice problem 6.1.P1: A projection coefficient and its scaled vector

Compute $a \cdot b = (-2)(-3) + 1(1) = 7$ and $a \cdot a = (-2)^2 + 1^2 = 5$. Thus

$$\frac{a \cdot b}{a \cdot a} = \frac{7}{5}, \qquad \left(\frac{a \cdot b}{a \cdot a}\right) a = \begin{bmatrix} -14/5 \\ 7/5 \end{bmatrix}.$$

The first answer is a scalar, while the second is a vector.

Practice problem 6.1.P2: Normalize the three-coordinate vector

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{4}{3})^2 + (-1)^2 + (\frac{2}{3})^2} = \frac{\sqrt{29}}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4\sqrt{29}}{29} \\ -\frac{3\sqrt{29}}{29} \\ \frac{2\sqrt{29}}{29} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Practice problem 6.1.P3: Check orthogonality directly

The inner product is $5(4/3) + 6(-1) + (-1)(2/3) = 20/3 - 6 - 2/3 = 0$. Therefore d is orthogonal to c; checking the zero inner product is exactly the definition needed.

Practice problem 6.1.P4: Orthogonality survives normalization

The unit vector u from Practice 2 is $c/\|c\|$. By linearity in the second argument, $d \cdot u = (1/\|c\|)(d \cdot c) = 0$ using Practice 3.

Conclusion. Multiplying one of two orthogonal vectors by a scalar does not destroy orthogonality.

Exercise 6.1.1: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$u \cdot u = (-1)(-1) + (2)(2) = 5,$$
$$v \cdot u = (4)(-1) + (6)(2) = 8.$$

Conclusion. The requested ratio is $(v \cdot u)/(u \cdot u) = \frac{8}{5}$.

Exercise 6.1.2: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$w \cdot w = (3)(3) + (-1)(-1) + (-5)(-5) = 35,$$
$$x \cdot w = (6)(3) + (-2)(-1) + (3)(-5) = 5.$$

Conclusion. The requested ratio is $(x \cdot w)/(w \cdot w) = \frac{1}{7}$.

Exercise 6.1.3: Inner-product and norm calculations

The squared length is $w \cdot w = 35$. Multiply every coordinate by its reciprocal:

$$\frac{1}{w \cdot w} w = \frac{1}{35} \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} \frac{3}{35} \\ -\frac{1}{35} \\ -\frac{1}{7} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.4: Inner-product and norm calculations

The squared length is $u \cdot u = 5$. Multiply every coordinate by its reciprocal:

$$\frac{1}{u \cdot u} u = \frac{1}{5} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.5: Inner-product and norm calculations

The numerator is $u \cdot v = 8$ and the denominator is $v \cdot v = 52$. Therefore

$$\left(\frac{u \cdot v}{v \cdot v}\right) v = \frac{2}{13} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{8}{13} \\ \frac{12}{13} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.6: Inner-product and norm calculations

The numerator is $w \cdot x = 5$ and the denominator is $x \cdot x = 49$. Therefore

$$\left(\frac{w \cdot x}{x \cdot x}\right) x = \frac{5}{49} \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{30}{49} \\ -\frac{10}{49} \\ \frac{15}{49} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.7: Inner-product and norm calculations

By the norm definition,

$$\|w\| = \sqrt{(3)^2 + (-1)^2 + (-5)^2} = \sqrt{35} = \sqrt{35}.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.8: Inner-product and norm calculations

By the norm definition,

$$\|x\| = \sqrt{(6)^2 + (-2)^2 + (3)^2} = \sqrt{49} = 7.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.9: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-30)^2 + (40)^2} = 50.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.10: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-6)^2 + (4)^2 + (-3)^2} = \sqrt{61}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{6\sqrt{61}}{61} \\ \frac{4\sqrt{61}}{61} \\ -\frac{3\sqrt{61}}{61} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.11: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{7}{4})^2 + (\frac{1}{2})^2 + (1)^2} = \frac{\sqrt{69}}{4}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{7\sqrt{69}}{69} \\ \frac{2\sqrt{69}}{69} \\ \frac{4\sqrt{69}}{69} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.12: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{8}{3})^2 + (2)^2} = \frac{10}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.13: Inner-product and norm calculations

Subtract the point coordinates first:

$$a - b = \begin{bmatrix} 10 \\ -3 \end{bmatrix} - \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \end{bmatrix}.$$

Conclusion. The distance is $\|a - b\| = \sqrt{125} = 5\sqrt{5}$. Reversing the subtraction gives the same length.

Exercise 6.1.14: Inner-product and norm calculations

Subtract the point coordinates first:

$$a - b = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} - \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ -6 \end{bmatrix}.$$

Conclusion. The distance is $\|a - b\| = \sqrt{68} = 2\sqrt{17}$. Reversing the subtraction gives the same length.

Exercise 6.1.15: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (8)(-2) + (-5)(-3) = -1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.16: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (12)(2) + (3)(-3) + (-5)(3) = 0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.17: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (3)(-4) + (2)(1) + (-5)(-2) + (0)(6) = 0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.18: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (-3)(1) + (7)(-8) + (4)(15) + (0)(-7) = 1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.19: Inner-product and norm calculations

(a) True. It is the squared-norm identity $\|v\|^2 = v \cdot v$.

(b) True. Scalar distributivity gives $u \cdot (cv) = c(u \cdot v)$.

(c) True. Squaring the equal distances and expanding gives $\|u\|^2 - 2u \cdot v + \|v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. Hence $4u \cdot v = 0$, so the vectors are orthogonal.

(d) False. Take $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Its column space contains $(1, 0)^T$ and its null space contains $(-1, 1)^T$; their dot product is -1, not zero. The correct general pairing is *row space* with null space.

(e) True. For $w = \sum_j c_j v_j$ we have $x \cdot w = \sum_j c_j (x \cdot v_j) = 0$; thus $x \in W^\perp$.

Exercise 6.1.20: Inner-product and norm calculations

(a) True. Symmetry gives $u \cdot v = v \cdot u$, so their difference is zero.

(b) False for negative scalars. For $c = -1$ and $v \neq 0$, $\|cv\| = \|v\|$ but $c\|v\| = -\|v\|$. The correct rule uses $|c|$.

(c) True. Being orthogonal to every member of W is the definition of membership in $W^\perp$.

(d) True. Expand $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. The given equality makes $2u \cdot v = 0$.

(e) True. $Ax = 0$ means every row has dot product zero with x. The same holds for every linear combination of rows, hence all of Row A.

Exercise 6.1.21: Inner-product and norm calculations

(b) Distribute using the transpose and matrix multiplication rules.

$$(u + v) \cdot w = (u + v)^T w = (u^T + v^T)w = u^T w + v^T w = u \cdot w + v \cdot w.$$

Here we used $(u + v)^T = u^T + v^T$ and distribution of matrix products across addition, both from Chapter 2.

(c) Move a scalar through the products.

The transpose rule $(cu)^T = cu^T$ and scalar associativity of products give

$$(cu) \cdot v = (cu)^T v = c(u^T v) = c(u \cdot v) = u^T (cv) = u \cdot (cv).$$

Each expression is a 1×1 product, identified with its real scalar value.

Exercise 6.1.22: Inner-product and norm calculations

We have $u \cdot u = u_1^2 + u_2^2 + u_3^2$. Every real square is nonnegative, so the sum is nonnegative. If the sum is zero, none of the nonnegative summands can be positive, so $u_i^2 = 0$ for every i. This forces all three entries to be zero. Conversely the zero vector has zero inner product with itself.

Conclusion. $u \cdot u \geq 0$, with equality if and only if $u = 0$.

Exercise 6.1.23: Inner-product and norm calculations

Compute directly rather than invoking the Pythagorean theorem:

$$u \cdot v = (2)(-7) + (-5)(-4) + (-1)(6) = 0,$$

$$\|u\|^2 = 30, \qquad \|v\|^2 = 101, \qquad u + v = \begin{bmatrix} -5 \\ -9 \\ 5 \end{bmatrix}.$$

$$\|u + v\|^2 = (-5)^2 + (-9)^2 + (5)^2 = 131 = 30 + 101.$$

The dot product is zero, so this example agrees with the orthogonal-sum identity, which has now been verified by arithmetic.

Exercise 6.1.24: Inner-product and norm calculations

Use bilinearity and symmetry to expand both squared norms:

$$\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2,$$

$$\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2.$$

Adding cancels the two cross terms and gives $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

Conclusion. The parallelogram law holds for all $u, v \in \mathbb{R}^n$; orthogonality is not assumed.

Exercise 6.1.25: Inner-product and norm calculations

Orthogonality to $(a, b)^T$ is the single homogeneous equation $ax + by = 0$. If $a = b = 0$, every (x, y) works, so $H = \mathbb{R}^2$. If $(a, b) \neq (0, 0)$, the nonzero vector $(-b, a)^T$ is orthogonal to $(a, b)^T$ and its multiples solve the equation. Conversely, if $a \neq 0$ solve $x = -(b/a)y$; if $a = 0$ then $b \neq 0$ and the equation gives $y = 0$. Either way every solution is a multiple of $(-b, a)^T$.

Conclusion. For $v \neq 0$, H is the line through the origin perpendicular to v; for $v = 0$, it is the entire plane.

Exercise 6.1.26: Inner-product and norm calculations

We can write $W = \text{Nul}[5\ -6\ 7]$. The Chapter 4 theorem that a null space is a subspace therefore applies. Explicitly its equation is $5x_1 - 6x_2 + 7x_3 = 0$, one nonzero homogeneous linear constraint. Solving for x_1 leaves two freely chosen coordinates, so the geometric set is a plane through the origin. The normal direction is u because $u \cdot x = 0$ for every vector in the plane.

Conclusion. W is the plane perpendicular to $(5, -6, 7)^T$ through zero.

Exercise 6.1.27: Inner-product and norm calculations

By distributivity, $y \cdot (u + v) = y \cdot u + y \cdot v = 0 + 0 = 0$. The two zero values follow from the two given orthogonality assumptions.

Conclusion. y is orthogonal to $u + v$.

Exercise 6.1.28: Inner-product and norm calculations

Take any $w \in \text{Span}\{u, v\}$. There are real scalars c_1, c_2 with $w = c_1u + c_2v$. Then $y \cdot w = c_1(y \cdot u) + c_2(y \cdot v) = 0$. Because w was arbitrary, the claim holds for every vector in the span, not merely for its two generators.

Conclusion. $y \in (\text{Span}\{u, v\})^{\perp}$.

Exercise 6.1.29: Inner-product and norm calculations

An arbitrary $w \in W$ can be written $w = \sum_{j=1}^{p} c_j v_j$. Apply distributivity repeatedly:

$$x \cdot w = \sum_{j=1}^{p} c_j (x \cdot v_j) = \sum_{j=1}^{p} c_j 0 = 0.$$

Thus every w in W is orthogonal to x.

Conclusion. Testing orthogonality against a spanning family tests all of W.

Exercise 6.1.30: Inner-product and norm calculations

(a) Closure under scalar multiplication.

Let $z \in W^{\perp}$, $c \in \mathbb{R}$, and let u be any vector of W. Then $(cz) \cdot u = c(z \cdot u) = c0 = 0$. Since this holds for every $u \in W$, $cz \in W^{\perp}$.

(b) Closure under addition.

For $z_1, z_2 \in W^{\perp}$ and any $u \in W$, $(z_1 + z_2) \cdot u = z_1 \cdot u + z_2 \cdot u = 0 + 0 = 0$. Hence $z_1 + z_2 \in W^{\perp}$.

(c) Include zero and apply the subspace test.

The zero vector has zero dot product with every $u \in W$, so it lies in $W^{\perp}$. Together with (a) and (b), these are all three required subspace properties.

Conclusion. $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Exercise 6.1.31: Inner-product and norm calculations

If $x \in W$ and $x \in W^{\perp}$, then the latter membership makes x orthogonal to every member of W, including itself. Thus $x \cdot x = \|x\|^2 = 0$, which forces $x = 0$ by positive definiteness.

Conclusion. $W \cap W^{\perp} = \{0\}$.

Exercise 6.1.32: Inner-product and norm calculations

(a) Check all column lengths and mutual dot products.

The single exact calculation $A^T A = I_4$ states that every column has squared length 1 and all six off-diagonal pairwise products are zero. Thus the four lengths are $1, 1, 1, 1$ and the six requested dot products are 0.

(b)–(d) Perform three reproducible random experiments.

Using integer draws from $[-3, 3]$ with Python random seed 61032 gives the following three nonzero pairs. Listed equalities compare the original and transformed quantities.

$$u = \begin{bmatrix} 2 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{17}, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = -\frac{\sqrt{102}}{17}.$$

$$u = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{14}, \quad \|v\| = \|Av\| = \sqrt{14},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{5}{14}.$$

$$u = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 2 \\ -1 \\ -1 \end{bmatrix};$$

$$\|u\| = \|Au\| = 2, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{\sqrt{6}}{6}.$$

Prove the observed behavior for every pair.

For arbitrary u, v, $(Au)^T(Av) = u^T A^T Av = u^T v$. Taking $v = u$ proves norm preservation, and dividing by nonzero norms proves angle preservation for nonzero vectors.

Conclusion. A preserves lengths and angles; the numerical experiments are instances of this identity.

Exercise 6.1.33: Inner-product and norm calculations

Compute two actual random examples with one fixed nonzero direction.

Using seed 61033 and integer draws from $[-3, 3]$, take $v = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, with $v \cdot v = 5$. Each scalar below multiplies this same explicitly given column.

$$x = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 1 \end{bmatrix},$$

$$T(x) = (\frac{8}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (1)v, \quad T(10x) = (16)v.$$
$$x = \begin{bmatrix} -2 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} -1 \\ 2 \\ 3 \\ -1 \end{bmatrix},$$
$$T(x) = (-\frac{1}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (-\frac{4}{5})v, \quad T(10x) = (-2)v.$$

Verify the conjectured linearity algebraically.

For scalars a, b and any inputs x, y, distributivity gives

$$T(ax+by) = \frac{(ax+by)\cdot v}{v\cdot v}v = aT(x) + bT(y).$$

The fixed denominator is nonzero because $v \neq 0$.

Conclusion. Projection onto a fixed nonzero direction is linear; the displayed experiments satisfy additivity and scalar homogeneity exactly.

Exercise 6.1.34: Inner-product and norm calculations

Read the free variables from an exact row reduction.

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 5 & 0 & -\frac{1}{3} \\ 0 & 1 & 1 & 0 & -\frac{4}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are $1, 2, 4$, leaving free coordinates $3, 5$. A null-space basis, with the second vector scaled by 3 to avoid fractions, and a nonzero-row basis for the row space give

$$N = \begin{bmatrix} -5 & 1 \\ -1 & 4 \\ 1 & 0 \\ 0 & -1 \\ 0 & 3 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 0 & 15 & 0 & -1 \\ 0 & 3 & 3 & 0 & -4 \\ 0 & 0 & 0 & 3 & 1 \end{bmatrix}.$$

The two columns of N have independent free-coordinate choices, and the three rows of R are scaled nonzero rows of RREF, so they are bases of the required spaces.

Multiply the two basis matrices.

$$RN = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Each entry is the dot product of one row-space basis vector with one null-space basis vector. All are zero, illustrating $(\text{Row } A)^{\perp} = \text{Nul } A$ from Theorem 3. The additional check $AN = 0$ verifies the null-space computation directly.